

HEALTHCONNECT DATA INTELLIGENCE

Healthcare Data Warehouse & Analytics

KPI REPORT

KPI Results and Validation Evidence

Prepared for Institute Submission

KPI Overview

This report presents the output of the 11 KPI views developed for the HealthConnect Data Intelligence project. The values below are transcribed from the SQL Server Management Studio result screenshots supplied for this report.

KPI 01 – Yearly Claim Trend

View: vw_KPI01_Yearly_Claim_Trend

Purpose: Yearly total claims, total claim amount, and average claim amount paid.

Output captured from the supplied SSMS screenshot:

Claim_Year	Total_Claims	Total_Claim_Amount	Average_Claim_Amount_Paid
2020	4260	116816260.83	16975.9600821596
2021	4334	120478501.26	17224.7173557915
2022	4259	117233965.11	17001.7165085701
2023	4460	123071460.10	17103.8077331838
2024	4423	121082764.95	16988.2963463713
2025	3264	90173832.6100001	17075.7325091911

KPI 02 – Payer Claims

View: vw_KPI02_Payer_Claims

Purpose: Total claims and total paid claim amount by payer.

Output captured from the supplied SSMS screenshot:

payer_id	payer_name	Total_Claims	Total_Paid_Claim_Amount
6	SBI GENERAL INSURANCE	2566	43832592.36
9	RELIANCE HEALTH INSURANCE	2584	43536576.48
3	HDFC ERGO HEALTH INSURANCE	2537	43421009.4300001
5	ORIENTAL INSURANCE	2518	43035954.86
4	NEW INDIA ASSURANCE	2502	42757500.9399999
1	STAR HEALTH INSURANCE	2469	42657262.2
8	MAX BUPA	2457	42236382.16
10	UNITED INDIA INSURANCE	2468	41960837.3899999
7	BAJAJ ALLIANZ	2479	41903761.3
2	ICICI LOMBARD	2420	41195356.6

KPI 03 – Claim Amount by Period

View: vw_KPI03_Claim_Amount_Period

Purpose: Current-year and previous-year YTD, QTD, MTD, and WTD claim amounts.

Output captured from the supplied SSMS screenshot:

Period_Type	YTD_Total_Claim_Amount	QTD_Total_Claim_Amount	MTD_Total_Claim_Amount	WTD_Total_Claim_Amount
Current Year	90173832.6100001	2055612.89	2055612.89	2717366.28
Previous Year	92372204.06	1337843.63	1337843.63	1968617.86

KPI 04 – Provider Patient Engagement

View: vw_KPI04_Provider_Patient_Engagement

Purpose: Provider engagement ranking with yearly visit counts over the five-year analysis period.

Output captured from the supplied SSMS screenshot:

Rank	Provider ID	Provider	Visit Year	Total Visits	Five-Year Total
1	77	LAKSHMI MEHTA	2021	25	121
1	77	LAKSHMI MEHTA	2022	23	121
1	77	LAKSHMI MEHTA	2023	26	121
1	77	LAKSHMI MEHTA	2024	29	121
1	77	LAKSHMI MEHTA	2025	18	121
2	80	ASHA BANERJEE	2021	23	118
2	80	ASHA BANERJEE	2022	19	118
2	80	ASHA BANERJEE	2023	28	118
2	80	ASHA BANERJEE	2024	32	118
2	80	ASHA BANERJEE	2025	16	118
3	17	ISHAAN MALHOTRA	2021	24	112
3	17	ISHAAN MALHOTRA	2022	27	112
3	17	ISHAAN MALHOTRA	2023	26	112
3	17	ISHAAN MALHOTRA	2024	20	112
3	17	ISHAAN MALHOTRA	2025	15	112

Note: The supplied screenshot displayed more rows than are reproduced in this report table; the table above records the visible leading results.

KPI 05 – Diagnosis Claim Cost

View: vw_KPI05_Diagnosis_Claim_Cost

Purpose: Claim count and total paid claim amount associated with diagnoses. The screenshot shows the highest-cost diagnosis rows.

Output captured from the supplied SSMS screenshot:

diagnosis_id	diagnosis_description	Total_Claims	Total_Paid_Claim_Amount
7974	HEMOPHILIA	9	243506.05
7976	LEUKEMIA	9	243506.05
7975	ULCERATIVE COLITIS	9	243506.05
9386	MULTIPLE SCLEROSIS	12	229240.07
71	PNEUMONIA	10	194315.16
9853	PANCREATITIS	8	191302.83
9852	WHOOPING COUGH	8	191302.83
3561	YELLOW FEVER	8	187830.94
3560	CHRONIC KIDNEY DISEASE	8	187830.94
5678	ANEMIA	8	187483.62
5679	POLIO	8	187483.62
936	ANTHRAX	9	187137.32
938	SCHIZOPHRENIA	9	187137.32
937	STROKE VIRUS	9	187137.32
495	EBOLA VIRUS	7	185390.34

Note: The supplied screenshot displayed more rows than are reproduced in this report table; the table above records the visible leading results.

KPI 06 – Yearly Visits and Claim Cost

View: vw_KPI06_Yearly_Visit_Claim_Cost

Purpose: Yearly total patient visits, total claim cost, and average cost per visit.

Output captured from the supplied SSMS screenshot:

Visit_Year	Total_Patient_Visits	Total_Claim_Cost	Average_Cost_Per_Visit
2020	380	20675990.35	54410.50
2021	1537	84240142.6000001	54808.16
2022	1521	85396367.43	56144.88
2023	1544	88896982.3400001	57577.51
2024	1506	83582766.0499999	55499.84
2025	1159	63742284.9499999	54997.66

KPI 07 – Gender Claim Analysis

View: vw_KPI07_Gender_Claim_Analysis

Purpose: Total claims, total claim amount, and average claim amount by patient gender.

Output captured from the supplied SSMS screenshot:

Gender	Total_Claims	Total_Claim_Amount	Average_Claim_Amount
F	11920	201912496.15	16938.97
M	13080	224624737.57	17173.15

KPI 08 – Most Prescribed Medications

View: vw_KPI08_Most_Prescribed_Medications

Purpose: Medication frequency based on prescription count. The screenshot displays the highest-frequency medications.

Output captured from the supplied SSMS screenshot:

drug_name	Prescription_Count
CEFIXIME	1049
OMEPRAZOLE	1041
LOSARTAN	1029
INSULIN	1005
CLOPIDOGREL	1000
ATORVASTATIN	1000
PARACETAMOL	997
LEVOFLOXACIN	993
METFORMIN	982
ASPIRIN	979
CAPTOPRIL	978
AMOXICILLIN	947

KPI 09 – Age Group Claims

View: vw_KPI09_Age_Group_Claims

Purpose: Total claims and total claim amount by patient age group.

Output captured from the supplied SSMS screenshot:

Age_Group	Total_Claims	Total_Claim_Amount
61+	6672	113830340.65
31-45	4546	77933204.2599998
46-60	5041	85820210.3600001
18-30	3853	65740651.96
0-17	4888	83212826.4900002

KPI 10 – Inpatient Stay YoY

View: vw_KPI10_Inpatient_Stay_YoY

Purpose: Average inpatient stay duration and percentage change compared with the previous year.

Output captured from the supplied SSMS screenshot:

Stay_Year	Average_Stay_Days	Previous_Year_Avg_Stay	YoY_Percentage_Change
2020	5.60	NULL	NULL
2021	5.58	5.60	-0.39
2022	5.52	5.58	-1.01
2023	5.65	5.52	2.27
2024	5.49	5.65	-2.77
2025	5.52	5.49	0.49

KPI 11 – Paid Claim YoY

View: vw_KPI11_Paid_Claim_YoY

Purpose: Total paid claim amount, previous-year amount, amount change, and YoY growth.

Output captured from the supplied SSMS screenshot:

Claim_Year	Total_Paid_Amount	Previous_Year_Amount	Amount_Change	YoY_Growth_Percentage
2020	72317589.95	NULL	NULL	NULL
2021	74651925.0200001	72317589.95	2334335.07000001	3.23
2022	72410310.6099999	74651925.0200001	-2241614.41000022	-3.00
2023	76282982.4899999	72410310.6099999	3872671.87999991	5.35
2024	75139234.74	76282982.4899999	-1143747.74999982	-1.50
2025	55735190.9099999	75139234.74	-19404043.83000002	-25.82

Testing & Validation Evidence

The following evidence should be retained with the project submission. The SQL implementation and validation scripts are included separately in the project ZIP.

- ☐ RAW → CLEANSED → REFINED row-count validation
- ☐ Data-quality validation: duplicates, NULLs, and orphan foreign-key references
- ☐ Pipeline_Log showing successful execution of the master pipeline
- ☐ Windows Task Scheduler configuration for the daily 10:00 PM IST run
- ☐ KPI validation queries and SSMS output screenshots

Conclusion

The HealthConnect Data Intelligence project provides a layered healthcare data warehouse pipeline from RAW through CLEANSED to REFINED, with stored-procedure orchestration, SCD Type 1 processing, pipeline logging, scheduled execution, and 11 reporting KPIs.